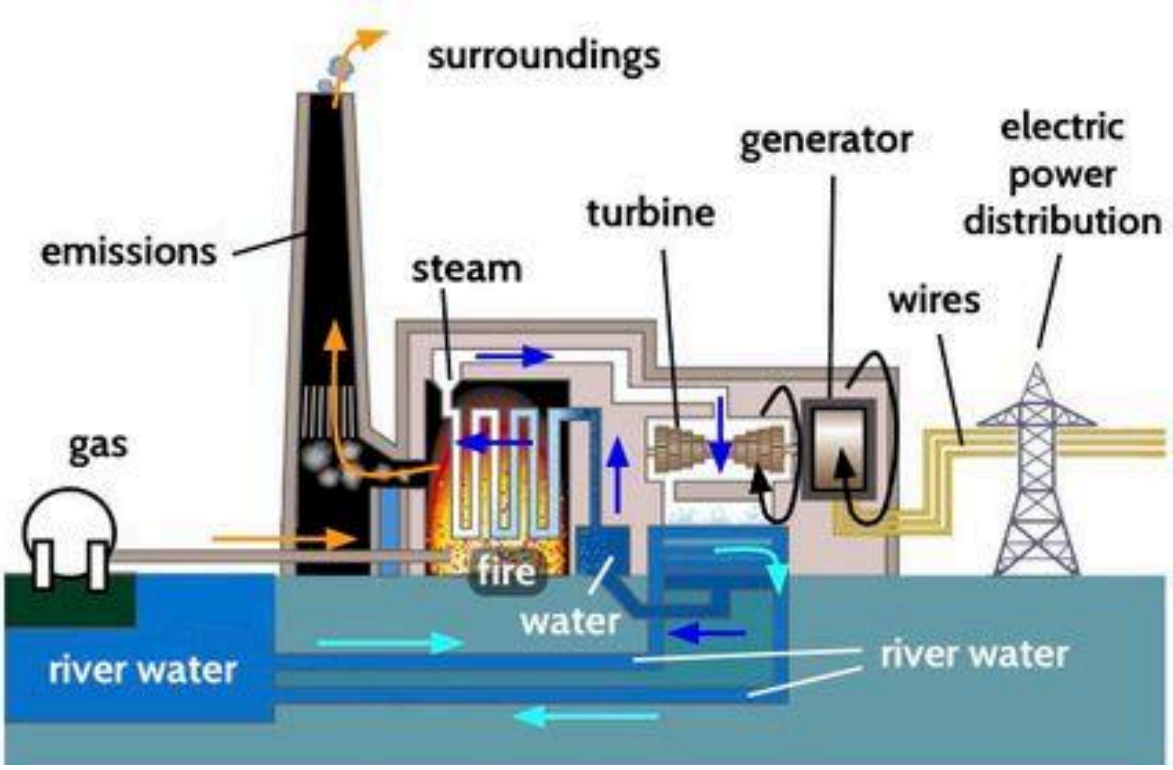
NAME: _____ **TEACHER:** _____

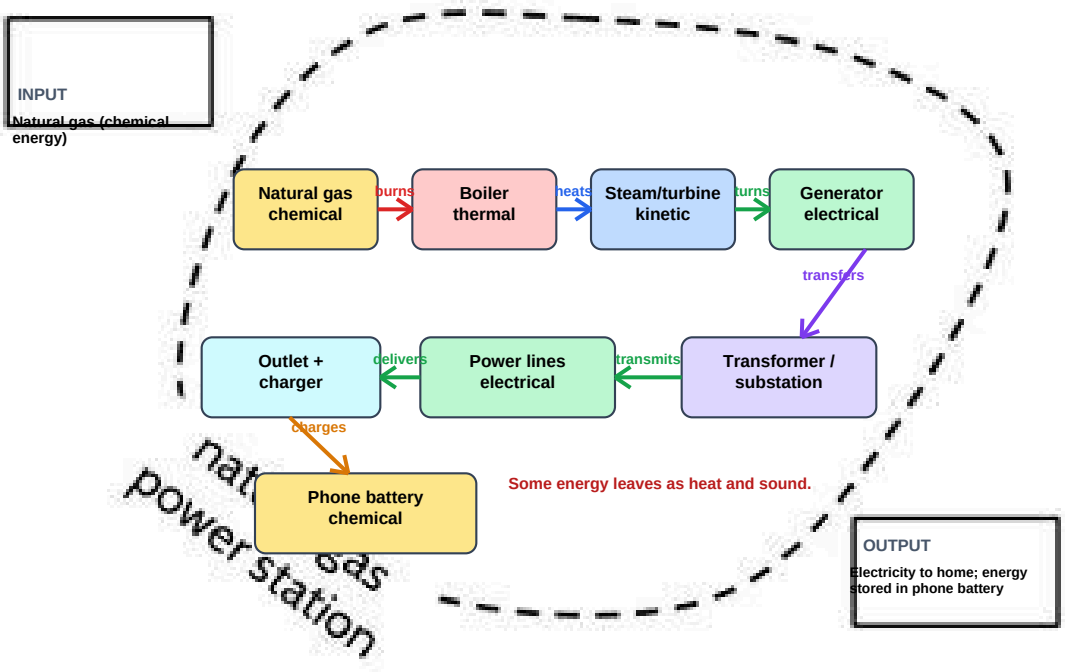
PERIOD:_____

Learning Target 0.1: I can develop an energy-transfer model to clearly and accurately explain components, interactions, and forms of energy in familiar and unfamiliar real-world systems.

Natural Gas Power Plant

This diagram shows key components in a **natural gas power plant**. Create an energy transfer model to show **how you charge your phone using natural gas.**





Be sure to include relevant vocabulary terms!

Part of Model	What do we need to communicate? What can we use?
Components *List what is included in your model on the back*	☑ Source of chemical energy ☑ Connections between objects ☑ Substations (*What is a substation?* A facility that changes voltage and routes electricity.) ☑ Objects using energy ☑ All relevant components are labeled ☐ Zoom-in bubble to show microscopic components, if needed *What did we use to represent these components?* Labeled boxes/icons represent natural gas, boiler, steam, turbine, generator, substation, power lines, outlet, charger, and phone battery.
Interactions *List what is included in your model on the back*	*How do the different components interact? What is connected to what? How does energy transfer? How much energy is transferred?* Gas burns in the boiler: chemical -> thermal energy. Steam spins the turbine: thermal -> kinetic. The turbine turns the generator: kinetic -> electrical. Electricity moves through the substation and power lines to the outlet and charger, then is stored as chemical energy in the phone battery. Some energy leaves as heat and sound. *What did we use to demonstrate interactions?* ☑ What process(es) are happening between objects ☑ **Quantity of energy generated/used is clearly shown and labeled** ☑ Energy transfer labeled with squiggly arrows (↝) ☑ Forces shown with solid arrows (⟶) ☑ Motion shown with dotted arrows (⇢) ☑ Direction of energy transfer is clear and labeled with a **verb** ☑ Label where energy enters/leaves the system ☐ Unknown interactions are indicated and labeled “magic”
Forms of Energy *List what is included in your model on the back*	*What forms of energy have we dealt with so far?* Chemical, thermal (heat), kinetic, and electrical energy. *How can you clearly differentiate forms of energy?* ☑ Forms of energy (chemical, thermal, kinetic, etc…) are drawn and labeled Use different colors plus a key; label each component and arrow with its energy form.
Does your model answer the prompt?	*What questions have we modeled answers to? How did these models directly answer the questions?* ☑ It is **clear** to the reader what my model is trying to communicate without additional explanation ☑ My model directly answers the prompt How does energy in natural gas become energy stored in a phone? The model traces every component, interaction, transformation, transfer direction, and energy loss from the plant to the battery.